

Guanyu Liu

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Research Interests: Reliable and Efficient AI; LLM Reasoning and Post-Training; Self-Evolving Agents

EDUCATION

University of Macau

Macau, China

B.S. Candidate in Statistics and Data Science GPA: 3.93/4.0

2023.08 - 2027.06

Relevant Coursework: *Data Structures and Object-Oriented Programming, Linear Algebra, Probability Theory, Discrete Structures, Machine Learning, Pattern Recognition*

Honors & Scholarships

- IET Prize 2025 — Sole Recipient at University of Macau
- Hong Tai Day Scholarship (2024–2025) — Sole Recipient across 10 Universities
- Bank of China Scholarship (2023–2024) — One of Two Recipients at University of Macau

RESEARCH

MetaGate: Reliable Adaptive Test-Time Compute (Accepted at PRICAI 2026)

Macau, China

Independent Researcher and Sole Author

2026.03-2026.04

- Conducted comprehensive research on multiple language models, covering method design, implementation, large-scale evaluation, and statistical analysis, to discover their reliability in resolving difficult problems.
- Benchmarked Qwen3 models from 1.7B to 30B on MATH-500 and identified that uniform compute allocation outperformed a common difficulty signal in 13 of 20 cross-model transfers.
- Developed MetaGate, a target-label-free fallback to standardize ANLL and four-sample answer-consistency signals using frozen source statistics, measure their discrepancy, and apply adaptive allocation under designed circumstances (meta-confidence exceeds a fixed threshold).
- Reduced below-uniform transfers from 13 of 20 to 1 of 20 at 6.99 decoded samples per problem, averaging 0.28 percentage points above uniform and 0.79 points above a sample-budget-matched random fallback.

CARE: LLM-Guided Reaction Optimization

Montreal, QC

Research Assistant and Co-First Author (Advisor: Dr. Tianyu Shi of McGill University)

2026.05 - 2026.08

- Co-developed an LLM-guided system to facilitate selecting promising reaction experiments under a fixed budget by learning the executable scoring rules from prior laboratory outcomes.
- Designed and implemented a reference-conditioned controller to separate program synthesis from experiment selection.
- Implemented outcome-guided program evolution by recording each intervention decision and its supporting evidence before outcome reveal, then updating persistent controller state with the observed result to retain, revise, or regenerate the active scoring program.
- Made LLM proposals executable through compilation, restricted execution, complete finite-scoring, and candidate-order invariance checks; refined valid alternatives through budgeted exploitation/exploration and a reference-conditioned gate, falling back deterministically to the numerical reference on failure. Ranked first across eight tasks with 30 matched seeds each in normalized regret, best-so-far AUC, and Top-1% Success@15.

IPU: Label-Free Prompt Selection

Hangzhou, China

Visiting Student, MAPLE Lab (Advisor: Prof. Guojun Qi of Westlake University)

2026.06 - 2026.08

- Introduced a novel prompt selection method under distribution shift without target labels by leading the project from problem formulation and theoretical development through implementation and evaluation.
- Designed IPU-PromptWeighted, a source-calibrated answer-class posterior to evaluate response classes of each prompt instead of relying on raw vote counts.
- Applied source-only, item-grouped nested cross-fitting and multinomial proper loss to select the posterior without target-label tuning, achieving Spearman $\rho = 0.7208$, outperforming Mean Pairwise Agreement (0.5713), ZMV-style majority pseudo-labeling (0.5583), and direct source accuracy (0.4721) On a 40-prompt GSM8K pool, and a low-budget Equal-z fusion of IPU and agreement retained $\rho = 0.6802$ with 80% fewer target calls.
- Currently working on HotPotQA validation for an ICLR 2027 submission.

RESEARCH (CONTINUED)

Lean2Isabelle: Cross-Assistant Proof Translation

Macau, China

Research Assistant (Advisor: Prof. Derek Wong of University of Macau)

2025.04 - 2026.05

- Contributed as co-first author to the design and evaluation of a system to achieve accurate conversion of formal proofs from Lean into Isabelle from idea formulation through manuscript preparation.
- Developed the entire data pipeline with Herald-ISA (26,104 verified pairs), LeanWorkbook-ISA (9,683 RL instances), and MiniF2F-DSP (217 transfer cases).
- Implemented and fine-tuned Statement-SFT and Theory-SFT by introducing GRPO for theory generation, Minimal Valid Progress shaping, a difficulty-ordered curriculum, and Isabelle/PISA verifier rewards, achieving correct statement translation on 86.0% and 85.7% of two benchmarks and improved semantic proof success from 49.8% to 54.3%.
- Published at the ICML 2026 AI for Math Workshop; extended version under review at NeurIPS 2026.

Target-Label-Free Mitochondrial Segmentation

Macau, China

Research Assistant (Advisor: Prof. Long Chen of University of Macau)

2025.05 - 2026.08

- Developed a target-label-free domain adaptation pipeline, transferring target acquisition appearance to labeled source stacks while preserving source morphology, and avoiding error-prone target pseudo-labels.
- Designed a Semantic Readiness Controller to activate boundary-safe replay by assessing EMA representation drift, target image-only loss, and source-loss gains alongside MAD-estimated noise.
- Innovatively implemented a full-stack posterior-stability gate to prevent premature replay, achieving 90.85% F1 and 83.29% IoU on VNC III-to-Lucchi2—0.14 and 0.23 points above the fixed-timing baseline—with fewer total iterations for selected replay at iteration 2k rather than the fixed 4k schedule.
- Preparing an ICASSP 2027 submission.

PUBLICATIONS

Peer-Reviewed and Accepted Publications

- [1] **Liu, G.** (2026). “When Difficulty Signals Are Unreliable: Auditing and Gating Sample-Count Adaptive Test-Time Compute.” *PRICAI 2026*. Accepted regular paper; sole author.
- [2] **Liu, G.***, Ma, J.*, & Wong, D. F. (2026). “Lean2Isabelle: Factorized Cross-Assistant Proof Translation.” *The 3rd AI for Math Workshop at ICML 2026*. Workshop paper; extended version under review at NeurIPS 2026.
*Equal contribution.

Manuscripts Under Review

- [3] **Liu, G.***, Kong, W.*, Tang, C.*, Wang, Z., Zhang, B., Li, B., Zhang, P., & Shi, T. (2026). “CARE: Context-Aware Ranking Evolution with Executable Scoring Programs for Budgeted Reaction Optimization.” Under review at *AAAI 2027*. *Equal contribution.
- [4] Han, Y., Liu, B., Zhou, Z., **Liu, G.**, Zhang, Z., Yang, Y., Wang, W., Shi, I. N., Yunyan, Y., He, L., & Shi, T. (2026). “MAPGD: Multi-Agent Prompt Gradient Descent for Collaborative Prompt Optimization.” Under review at *Transactions on Machine Learning Research (TMLR)*; preliminary version presented at the *NeurIPS 2025 Workshop on Scaling Environments for Agents (SEA)*.

TECHNICAL SKILLS

Programming: Python, C++, Bash

LLM Training & Post-Training: PyTorch, Hugging Face Transformers, TRL

LLM Programming & APIs: DSPy, LiteLLM, OpenAI-compatible APIs

Inference & Infrastructure: vLLM, Slurm, Git, Linux, LaTeX